

From arguments *to agreement.*

Designing, measuring and repairing an annotation quality pipeline:
600 support tickets, two labeling rounds, three annotators,
and one evidence-driven guideline revision — the full account.

600	support tickets labeled (2 rounds × 3 personas · intent 8-class + ordinal sentiment)
0.8345 → 0.9554	Fleiss' κ on intent, before/after the v2.0 guideline revision
1.000 / 0.958	final-label accuracy vs the gold set (intent / sentiment)
4.7%	round-2 rework rate — vs the 15-20% industry ceiling
100%	tier-3 audit pass (n = 117, stratified)

PROGRAM WINDOW
2026-06-29 → 2026-08-24 · 2 rounds + calibration

ARTIFACTS
guidelines v1/v2 + changelog · labeled data · agreement notebook · QA audit · dashboard (print form of §§01-08)

SIMULATED-ANNOTATOR STUDY

All personas are documented, seeded simulation engines; every statistic is recomputed from raw labels in the repository — nothing in this report is typed by hand.

Executive summary

GuidelineForge is a complete, instrumented annotation program: real guidelines (v1 → v2), 600 customer-support tickets labeled for intent and sentiment, a 180-ticket blind triple-overlap for inter-annotator agreement, an embedded 72-item gold set, and a three-tier QA review (annotator → senior → auditor). Its purpose was to answer one question: can a measured disagreement-analysis loop turn mediocre label agreement into reliable, gold-verified label quality — and what does doing so teach us about running an annotation program?

- **Round 1 exposed hidden failure behind a passable average.**
Overall intent agreement (Fleiss' $\kappa = 0.83$) looked acceptable, but slice-level analysis found mixed-intent tickets at chance level ($\kappa = 0.03$), sarcasm sentiment at 30% unanimity, and cancellation at per-class $\kappa = 0.58$ — the worst class.
- **Agreement is not correctness.**
On anger-framed action requests, annotators were 75% unanimous yet only 25% correct vs. the gold set — v1's 'tone-first' rule was systematically mis-routing refunds to complaints. The gold set, not IAA, caught it.
- **The guideline was the defect, not the annotators.**
v2.0 replaced 'first-mention' tie-breaking with a precedence hierarchy, deleted the tone-first rule, and added sarcasm, polite-failure and short-fragment rules. Result: intent $\kappa\ 0.83 \rightarrow 0.96$, ordinal α on sentiment $0.61 \rightarrow 0.89$, bootstrap CIs disjoint.
- **The quality economics held.**
Rework settled at 1.7% (intent) / 3.3% (sentiment) against a 15–20% industry ceiling; the tier-3 audit passed 100% of 117 spot-checks; final label accuracy vs. gold reached 1.00 / 0.96.

metric	round 1 (v1.0)	round 2 (v2.0)	target
Fleiss' κ — intent	0.835	0.955	≥ 0.80 ✓
Fleiss' κ — sentiment	0.668	0.878	≥ 0.80 ✓
Krippendorff α — sentiment (ordinal)	0.610	0.885	≥ 0.80 ✓
mean Cohen's κ — intent	0.835	0.956	—
gold accuracy — intent (best annotator)	0.889	0.986	≥ 0.90
rework rate (either task)	—	0.047	$< 15\text{--}20\%$ ✓
tier-3 audit pass rate	—	1.000	≥ 0.95 ✓

A one-line version: after measuring where and **why** annotators disagreed, a targeted guideline revision lifted reliable agreement by +0.12 κ and final gold-verified accuracy to ~1.00 on intent — without replacing a single annotator.

Program design & data provenance

Corpus. 600 support tickets: 520 sampled from the public Bitext Customer Support LLM training set (26,872 rows, 27 fine-grained intents mapped to our 8-class taxonomy), plus 80 hand-written tickets in five *documented ambiguity families* — sarcasm, mixed intents, polite complaints, anger-framed action requests, and ultra-short fragments. ~115 organic rows carry designed sentiment wraps (documented augmentation; the raw Bitext corpus is almost entirely neutral). Ambiguity is deliberately engineered: agreement statistics are meaningless on trivially easy data.

Design. An annotation program's structure — who sees what, when — is its primary quality instrument. This program uses: a calibration subset (48 tickets) with a $\geq 85\%$ certification gate before production labeling; a blind overlap subset (180 tickets labeled by all three annotators) as the IAA measurement set; single-annotator production (420 tickets, reassigned between rounds to preserve blindness); and gold items (72, 12%) embedded invisibly in both streams for weekly accuracy tracking.

subset	n	labeled by	purpose
calibration	48	all 3, blind	certification gate + guideline test-bed
blind overlap	180	all 3, blind	IAA measurement (Cohen/Fleiss/Krippendorff)
production	420	1 primary	throughput + tier-2 review stream
gold set	72	(embedded)	accuracy vs ground truth, per annotator

Annotators (simulation disclosure). Three *documented, seeded* persona engines — a careful literalist, an empathetic reader, and a fast skimmer — label from observable text cues only; none ever see design or gold labels. Personas parameterize exactly the human factors that drive real disagreement: rule fidelity, reading depth, recency/skim bias, and sensitivity to emotional framing. The pipeline is persona-agnostic: drop real peer labels into data/for_peer_annotation/ and every statistic below recomputes unchanged (src/import_peer_labels.py).

Metric stack. Cohen's κ (pairs) · Fleiss' κ (3 raters) · Krippendorff's α (nominal intent; ordinal sentiment, which prices near-misses below extreme confusions) · per-class one-vs-rest κ · slice-level unanimity · bootstrap 95% CIs. All implemented from scratch and validated against scikit-learn, NLTK, and the canonical Fleiss (1971) worked example (0.210, matched to 3 decimals) — QA begins by calibrating the instruments.

Guideline evolution: v1.0 → v2.0

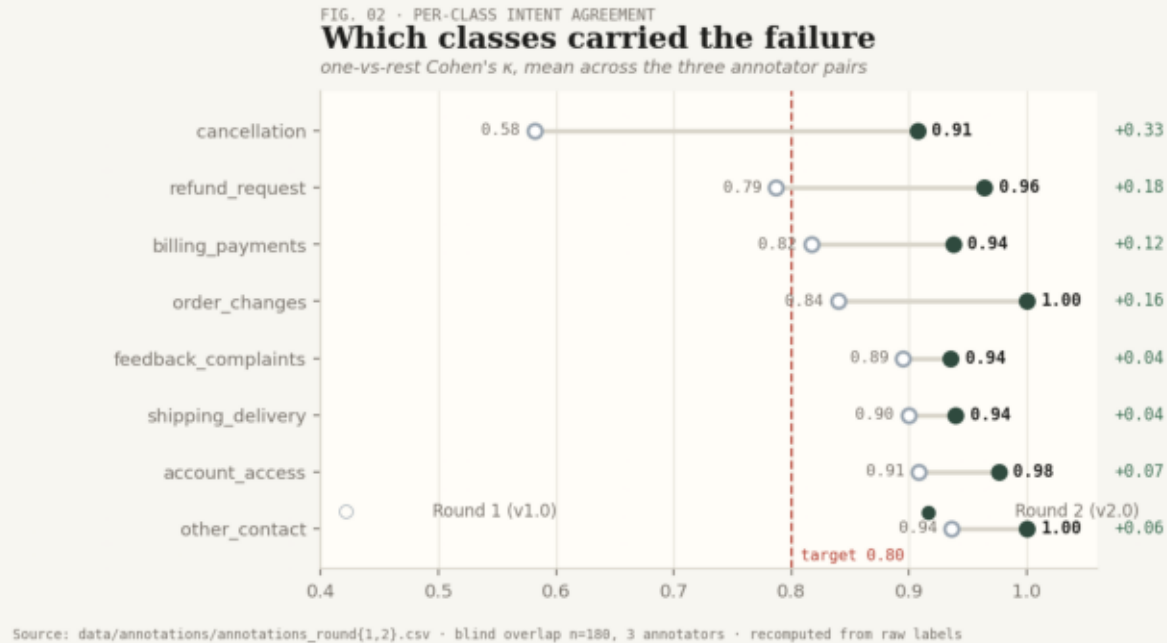
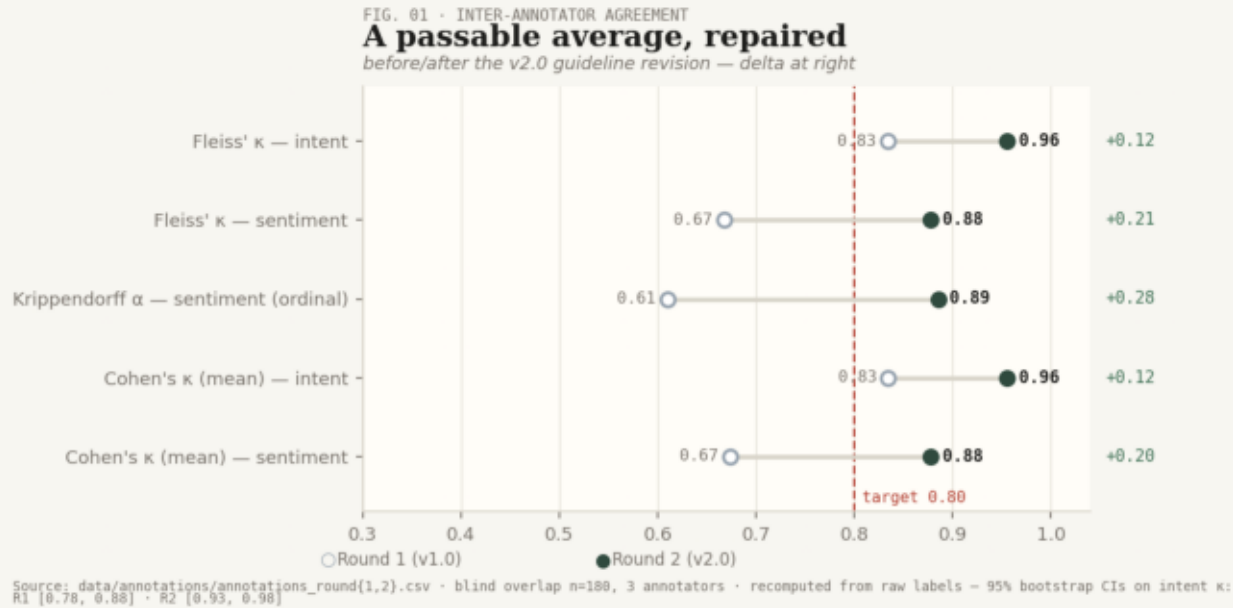
Disagreement analysis drove five targeted changes. Nothing about the taxonomy changed — the class boundaries were never the problem; the tie-breaking and tone rules were.

#	change	driving measurement (round 1)	effect (round 2)
C1	deleted tone-first rule R4	anger-framed requests: 75% unanimous, only 25% correct vs gold	gold intent acc 0.85–0.89 → 0.96–0.99
C2	intent precedence hierarchy	mixed-intent slice $\kappa = 0.03$, unanimous 12%	$\kappa\ 0.03 \rightarrow 0.87$, unanimous 12% → 94%
C3	sarcasm/contrast rule (R8)	sarcasm sentiment $\kappa = 0.25$, unanimous 30%	$\kappa\ 0.25 \rightarrow 0.57$, unanimous 30% → 80%
C4	polite-failure sentiment rule (R5)	polite slice sentiment $\kappa = 0.25$, unanimous 54%	$\kappa\ 0.25 \rightarrow 0.73$, unanimous 54% → 77%
C5	short-fragment rule (R9)	fragment sentiment $\kappa = -0.05$	$\kappa\ -0.05 \rightarrow 0.60$

The pattern worth memorizing: each change traces to a **measured** failure mode with a mechanistic explanation, and each re-measured after the fix. Guidelines iterate the same way code does — with a changelog, evidence, and regression checks.

Agreement: headline & per-class anatomy

Round 1 cleared the 0.80 bar on intent overall while failing on sentiment — and the per-class view shows exactly where: cancellation (0.58) and refund (0.79), the classes entangled by mixed-intent tickets.



The disagreement map: slices & mechanism

Slice-level unanimity (top) localizes failure to three language families; the annotator×annotator confusion matrix (bottom) names the mechanism — cancel≠refund≠billing tie-breaking plus skim errors.

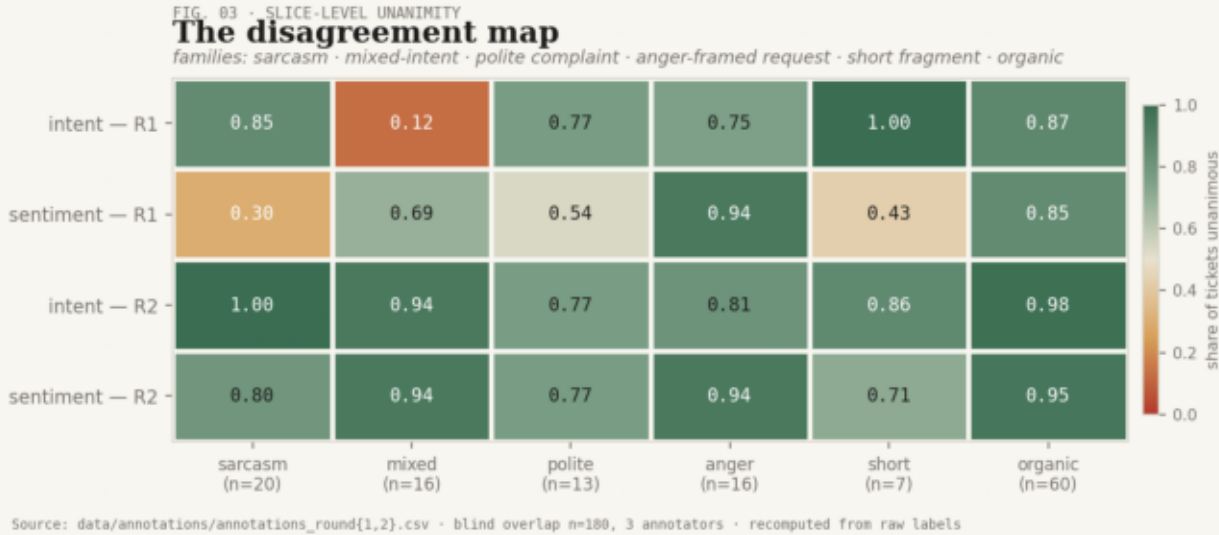
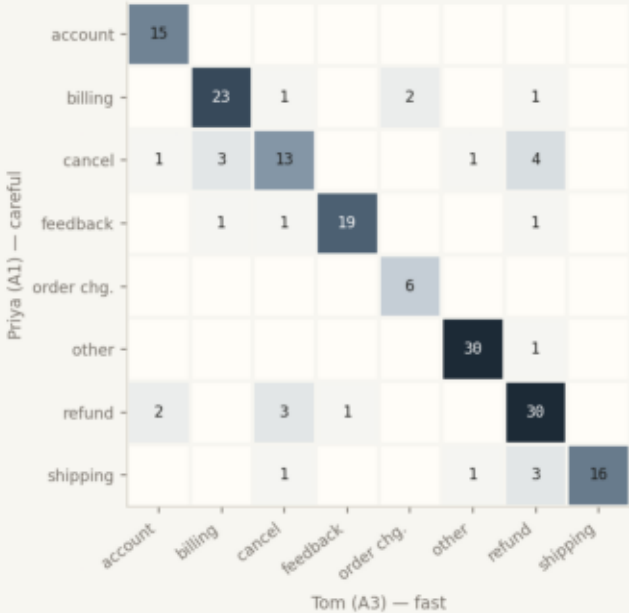


FIG. 06 · ANNOTATOR × ANNOTATOR, ROUND 1

Off-diagonal ink is the argument

rows: Priya (careful) · columns: Tom (fast) · n=180 blind overlap · cancel-refund-billing mass = guideline ambiguity · scattered singles = skim errors



Gold set: measuring correctness, not just consensus

Gold-set accuracy (top) is independent of IAA and catches what agreement cannot: unanimous-but-wrong rules. The weekly trend (bottom) shows the revision holding across all Round-2 batches.

FIG. 04 · GROUND-TRUTH ACCURACY

Consensus is not correctness

per-annotator accuracy on the embedded gold set — the QA floor at 0.90 is where retraining triggers

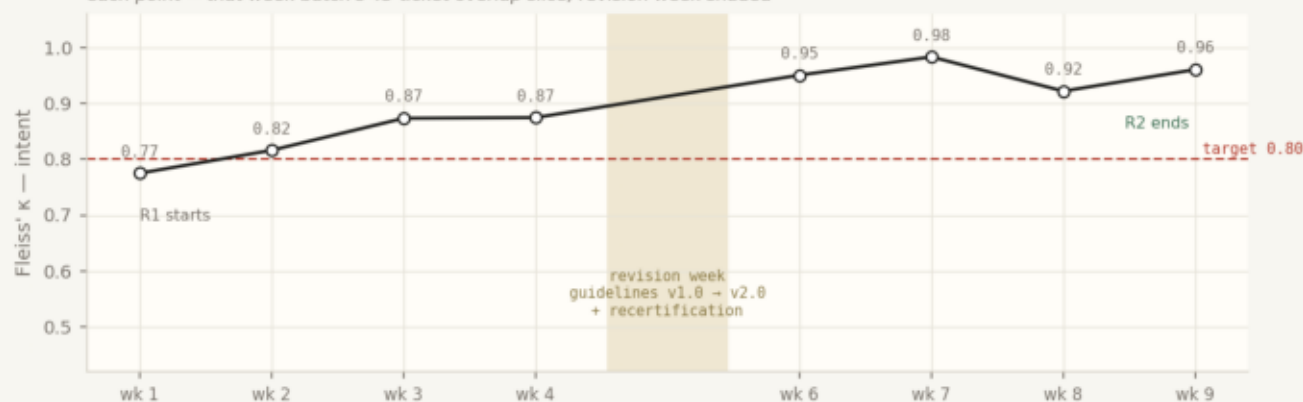


Source: data/gold_set.csv × annotations rounds 1-2 · annotators never saw gold flags

FIG. 05 · WEEKLY BATCHES

The revision held — every week after it

each point = that week batch's 45-ticket overlap slice; revision week shaded



Source: data/annotations/annotations_round{1,2}.csv · blind overlap $n=180$, 3 annotators · recomputed from raw labels

QA economics & throughput

Tier-2 rework by class (top-left) with the 15–20% industry ceiling for reference; the v1-era mislabel cost vs the v2 steady-state (top-right); throughput over the 9-week program (bottom).

FIG. 07 · QUALITY-ASSURANCE ECONOMICS
The steady state is boring — by design

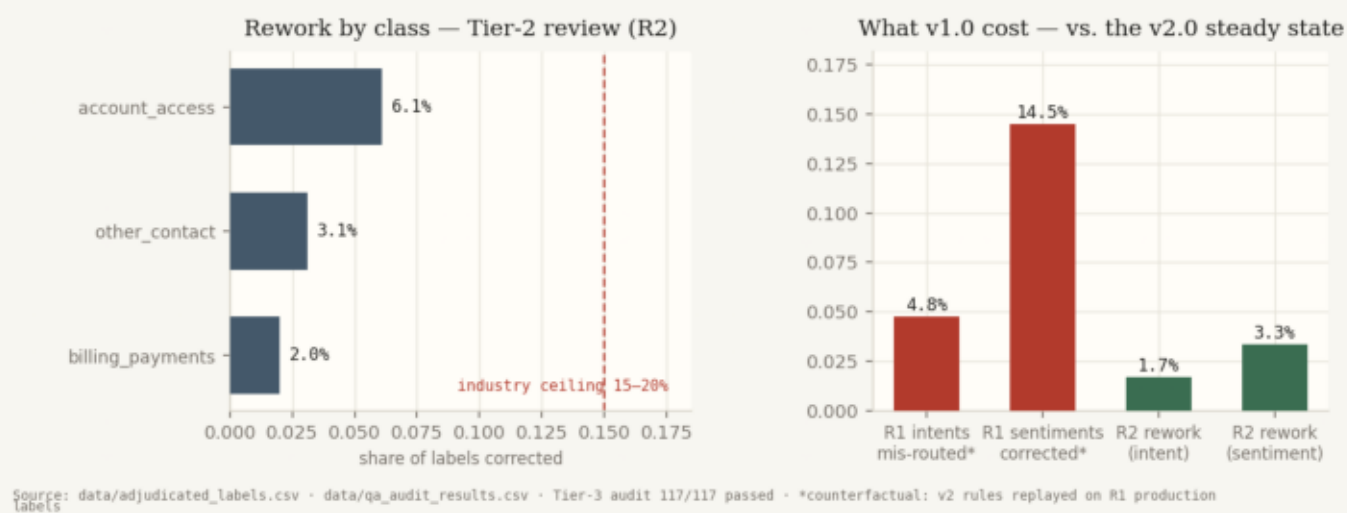
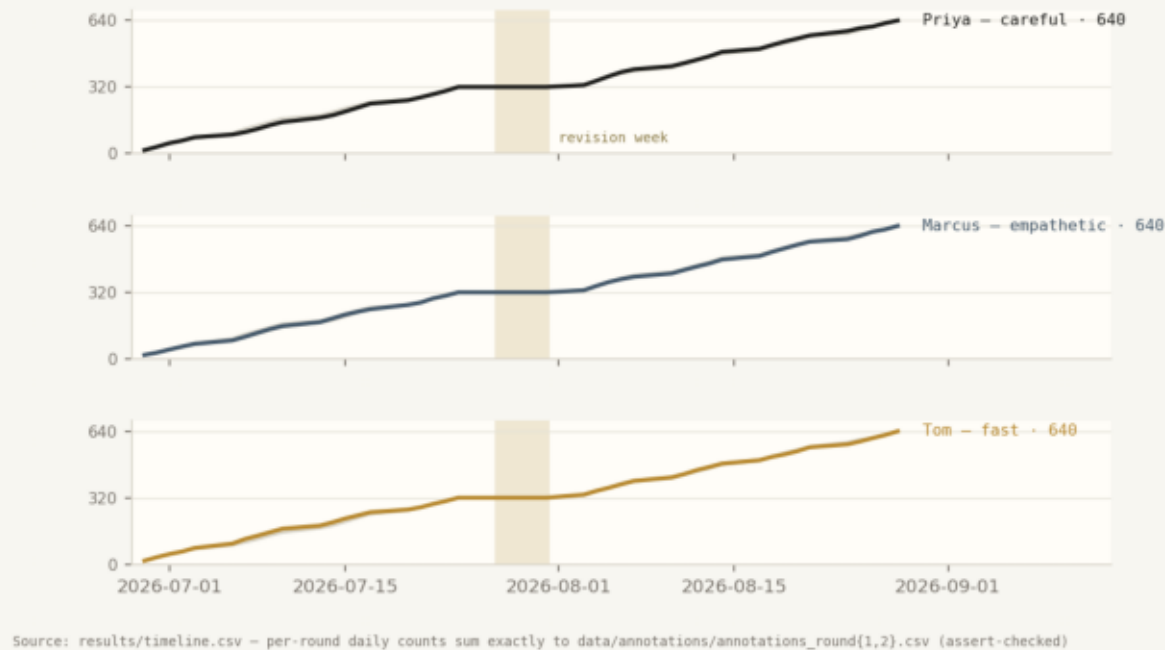


FIG. 08 · THROUGHPUT
Nine weeks, three workstations
cumulative labels per annotator (ghost lines = the other two) — totals reconcile 1:1 with the label files



Case study & limitations

The resume-ready version. Ran a 600-ticket, 3-annotator pilot on an 8-class intent + ordinal sentiment taxonomy; diagnosed guideline ambiguity behind chance-level agreement on mixed intents ($\kappa = 0.03$) and agreement- without-correctness on anger-framed requests (75% unanimous, 25% correct); revised the guidelines; and lifted Fleiss' κ from 0.83 to 0.96 on intent and ordinal Krippendorff's α from 0.61 to 0.89 on sentiment — finishing at 1.00 / 0.96 gold-verified final accuracy with a 4.7% rework rate against a 15–20% industry ceiling and a 100% tier-3 audit pass.

Known limitations (v2.1 backlog, logged from QA): · Statement-form policy questions ("i cannot check ur money back guarantee" — no '?', policy gate misses) stay mis-routed to refund_request. · Keyword-invisible typos ("acvount") fall to other_contact; a fuzzy-match pass or a label-studio text-hint would catch these. · Short-fragment sentiment remains the hardest slice (κ 0.60 — better, but below target): scheduled for worked-example expansion in v2.1. · Gold-set sentiment for organic rows rests on one expert pass; a second expert review round is the cheapest quality upgrade available to the program.

Simulation stance. Persona engines are transparent, seeded, documented — they exist so the full pipeline is reproducible and inspectable end-to-end. The methodology (overlap design, certification gates, gold embedding, tiered QA, slice-driven guideline iteration) transfers unchanged to live teams: swap persona labels for peer labels and every table re-computes.